

AI/ML Engineering & Architecture Training Roadmap

6-Week Hands-On Training Program

Program Overview

This program is designed for an experienced DevOps engineer transitioning into AI/ML, MLOps, LLMOps, and production AI engineering.

- 45 core credits / hours
 - 5 weeks × 9 hours = 45 hours
 - Week 6: lighter wrap-up, review, demo, discussion, and future direction
 - Week 6 also serves as a flexible buffer to revisit, reinforce, or complete any earlier topics that may need additional hands-on time
 - Hands-on assignments run alongside the main sessions
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Week 1

Local RAG Foundation

9 Credits / Hours

Focus

Build and understand a complete local RAG system.

Topics

- AI, ML, GenAI, LLM and RAG terminology
- RAG architecture
- PDF/TXT ingestion
- Text extraction and document chunking
- Embeddings and embedding models
- Similarity search

- Vector database
- Retrieval and context construction
- Local pretrained/open-source LLM
- Prompt engineering fundamentals
- Structured output
- Hallucination and grounding
- Basic RAG evaluation
- Simple API/UI

Outcome

Working local AI Knowledge Assistant over PDF/text documents.

Week 2

Cloud RAG, Evaluation & Automation

9 Credits / Hours

Focus

Move the RAG system to cloud and automate ingestion, evaluation, and deployment.

Topics

- Dockerizing the RAG application
- Cloud deployment and object storage
- Automated document ingestion and re-indexing
- Open-source vs hosted models
- Model selection
- Prompt, chunking, embedding and retrieval comparison
- Top-K and retrieval configuration
- RAG evaluation
- Evaluation/test datasets
- Automated regression testing
- Groundedness and hallucination checks
- Latency, cost, privacy and performance trade-offs
- CI/CD fundamentals
- Initial Harness integration

Outcome

Cloud-hosted RAG system with automated ingestion and measurable quality.

Week 3

Data Engineering & Classical ML Training

9 Credits / Hours

Focus

Prepare structured data and train custom ML models.

Topics

- Historical dataset collection/generation
- Data cleaning and missing data
- Dataset preparation
- Features and labels
- Train / validation / test split
- Supervised vs unsupervised learning
- Regression, classification and clustering
- Linear Regression
- Logistic Regression
- Decision Trees
- Random Forest
- XGBoost
- K-Nearest Neighbours
- Accuracy, precision, recall, F1 and confusion matrix
- Overfitting and underfitting
- Bias and variance
- Data leakage
- Class imbalance
- Model comparison

Outcome

Custom-trained ML model with evaluated and compared results.

Week 4

Deep Learning & MLOps

9 Credits / Hours

Focus

Understand neural-network training and production ML lifecycle management.

Topics

- Neural-network architecture
- Neurons and layers
- Activation functions
- Forward propagation
- Loss functions
- Backpropagation
- Optimizers
- Training loops
- Batch size and learning rate
- Model checkpoints
- CPU vs GPU training
- PyTorch fundamentals
- Experiment tracking
- MLflow
- Model artifacts and registry
- Model versioning
- Batch vs real-time inference
- Model serving
- Automated evaluation
- Retraining concepts
- CI/CD and Harness pipeline concepts

Outcome

Train, track, version, register and deploy a model through an MLOps-style lifecycle.

Week 5

Transformers, LLMs, Fine-Tuning & Harness

9 Credits / Hours

Focus

Understand modern LLMs, improve model quality, and automate model selection and release.

Topics

- NLP evolution
- Tokenization
- Embeddings
- Attention
- Transformer architecture
- Context windows
- Pre-training and inference
- Foundation models
- Open-source vs managed LLMs
- Model size, latency and cost trade-offs
- Model selection
- Prompting vs RAG vs fine-tuning
- Fine-tuning concepts
- Fine-tuning dataset preparation
- LoRA and PEFT
- Baseline vs improved model
- RAG/model/prompt comparison
- Data drift
- Concept drift
- Training-serving skew
- Production model degradation
- AI/ML evaluation harness
- Automated evaluation and regression testing
- Quality gates
- Harness CI/CD
- Approval, promotion and rollback
- Automated deployment of the best-performing version

Outcome

Compare, improve, fine-tune, evaluate and automatically release better AI/model versions.

Week 6

Wrap-Up & Final Demonstration

Focus

Consolidate the complete learning journey.

Topics

- Final project demonstration
- End-to-end architecture review
- RAG vs ML vs fine-tuning decisions
- Harness pipeline walkthrough
- Evaluation and improvement review
- Monitoring and security overview
- Agents and MCP introduction
- Production considerations
- Troubleshooting and discussion
- Assignment review
- Future learning roadmap

Outcome

Confidently explain and demonstrate the complete AI/ML lifecycle.
